

Problem

Determine the inverse Laplace transform of

$$F(s) = 5 + \frac{6}{s+4} - \frac{7s}{s^2+25}$$

Step-by-step solution

Step 1 of 1

The function $F(s)$ is defined as,

$$F(s) = 5 + \frac{6}{s+4} - \frac{7s}{s^2+25}$$

The inverse Laplace transform for $F(s)$ is,

$$\mathcal{L}^{-1}(F(s)) = \mathcal{L}^{-1}\left(5 + \frac{6}{s+4} - \frac{7s}{s^2+25}\right)$$

Apply linearity property to evaluate the inverse Laplace transform of $F(s)$.

$$\begin{aligned}\mathcal{L}^{-1}(F(s)) &= \mathcal{L}^{-1}(5) + \mathcal{L}^{-1}\left(\frac{6}{s+4}\right) - \mathcal{L}^{-1}\left(\frac{7s}{s^2+25}\right) \\ &= 5\mathcal{L}^{-1}(1) + 6\mathcal{L}^{-1}\left(\frac{1}{s+4}\right) - 7\mathcal{L}^{-1}\left(\frac{s}{s^2+5^2}\right)\end{aligned}$$

Consider the following Laplace transforms:

$$\mathcal{L}^{-1}(1) = \delta(t)$$

$$\mathcal{L}^{-1}\left(\frac{1}{s+4}\right) = e^{-4t}u(t)$$

$$\mathcal{L}^{-1}\left(\frac{s}{s^2+5^2}\right) = \cos 5t u(t)$$

Use the above Laplace transforms for finding the inverse Laplace transform for $F(s)$.

$$\begin{aligned}\mathcal{L}^{-1}(F(s)) &= 5\delta(t) + 6e^{-4t}u(t) - 7\cos(5t)u(t) \\ &= 5\delta(t) + (6e^{-4t} - 7\cos(5t))u(t)\end{aligned}$$

Therefore, the inverse Laplace transform of function $F(s)$ is $\boxed{5\delta(t) + (6e^{-4t} - 7\cos 5t)u(t)}$.